Column Differences and Alternating Antidiagonals of A260056

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Abstract

We prove both conjectures recorded in OEIS A260056. Every column is an explicit polynomial in the row index, of degree exactly one more than the column index. This proves the conjectured finite-difference recurrence and its minimal order. Alternating antidiagonal sums have a rational generating function that identifies them with the stated shift of A004524. All assertions hold at every row and column.

Let

$$T(n,k) = [x^k] \sum_{j=0}^{n} (1+x+x^2)^j \quad (n,k \geq 0),$$

with coefficients outside a row taken as zero. This is the polynomial definition of A260056 [1]. The entry conjectures that column k has vanishing differences of order $k+2$, and that its alternating antidiagonal sums equal A004524 shifted by three.

Theorem 1. *For every $n, k \geq 0$,*

$$T(n,k) = \sum_{i=\lceil k/2 \rceil}^{k} \binom{i}{k-i} \binom{n+1}{i+1}. \tag{1}$$

Consequently

$$\sum_{m=0}^{k+2} (-1)^m \binom{k+2}{m} T(n+m,k) = 0.$$

As a polynomial in n, column k has degree exactly $k+1$; no lower positive difference order vanishes identically.

Proof. Expand $(1+x+x^2)^j = \sum_{i=0}^{j} \binom{j}{i} x^i (1+x)^i$. Its coefficient of x^k is $\sum_i \binom{j}{i}\binom{i}{k-i}$, where only $\lceil k/2 \rceil \leq i \leq k$ contribute. Sum over j and apply $\sum_{j=0}^{n} \binom{j}{i} = \binom{n+1}{i+1}$ to obtain (1), including when $i > n$. The summand $i = k$ has degree $k+1$ and leading coefficient $1/(k+1)!$; every other summand has smaller degree. Each forward difference lowers the degree of a nonconstant polynomial by exactly one. Thus the $(k+2)$nd difference is zero and the $(k+1)$st difference is the nonzero constant one. The displayed signed form differs from the forward-difference convention only by the global sign $(-1)^{k+2}$. □

Theorem 2. *Define $c(m) = \sum_{k=0}^{m} (-1)^k T(m-k,k)$. Then for every $m \geq 0$,*

$$c(m) = \text{A004524}(m+3), \qquad c(4q+r) = 2q + \begin{cases} 1, & r = 0, \\ 2, & r = 1,2,3, \end{cases} \quad 0 \leq r < 4.$$

Proof. The row definition gives the bivariate generating function

$$F(t,x) = \sum_{n,k \geq 0} T(n,k) t^n x^k = \frac{1}{(1-t)(1-t(1+x+x^2))}.$$

Substitution $(t,x) = (z,-z)$ is legitimate as a formal series and weights each term by $(-1)^k z^{n+k}$. Hence

$$\sum_{m \geq 0} c(m) z^m = F(z,-z) = \frac{1}{(1-z)^2(1+z^2)}.$$

A004524 [2] has generating function $z^3/((1-z)^2(1+z^2))$ and initial coefficients zero through index two, proving the stated shift. Alternatively, the displayed four residue-class formulas have the same generating function, which verifies the identification without relying on a numerical comparison of terms. □

Validation and scope. The standard-library verifier independently constructs rows using trinomial convolution, checks the closed column formula through row 100 and column 40, audits the finite-difference identities, checks every antidiagonal through 100, and compares the complete published 100-row flattened table. The arguments prove the two conjectures for all indices; finite checks validate the implementation.

References

[1] W. Schulte, OEIS A260056, https://oeis.org/A260056, accessed October 26, 2026.

[2] The OEIS Foundation, A004524, https://oeis.org/A004524, accessed October 26, 2026.